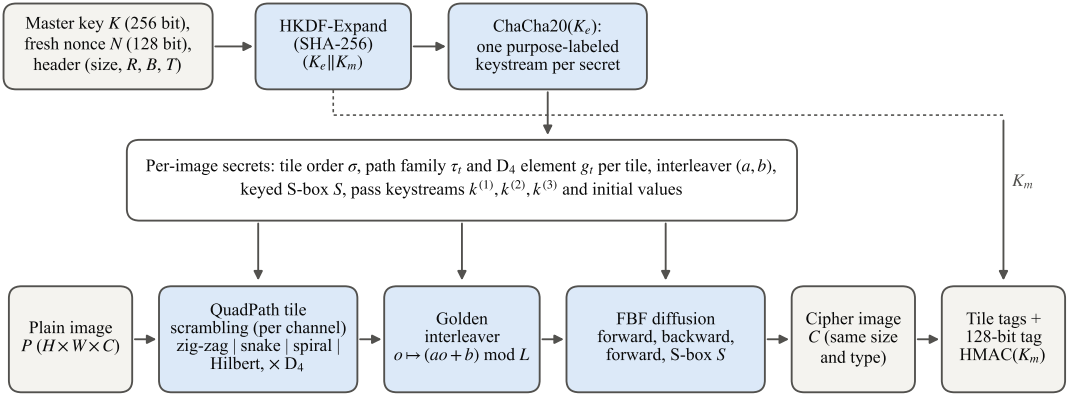
Master key K (256 bit), fresh nonce N (128 bit), header (size, R, B, T)
HKDF-Expand (SHA-256) ($K_e \| K_m$)
ChaCha20(K_e): one purpose-labeled keystream per secret
Per-image secrets: tile order σ, path family τ_t and D_4 element g_t per tile, interleaver (a, b), keyed S-box S, pass keystreams $k^{(1)}, k^{(2)}, k^{(3)}$ and initial values
K_m
Plain image P ($H \times W \times C$)
QuadPath tile scrambling (per channel) zig-zag | snake | spiral | Hilbert, $\times D_4$
Golden interleaver $o \mapsto (ao + b) \bmod L$
FBF diffusion forward, backward, forward, S-box S
Cipher image C (same size and type)
Tile tags + 128-bit tag HMAC(K_m)